

Practical Work Serie 3

Exercise 1

We consider the following algorithms:

```
Algorithm Algo_1 ;
  Var i,N : integer ;
  Nb : array [50] of integer ;
Begin
  Repeat
    Read (N);
  Until ((N>0) and (N≤50)) ;

  For (i ← 1 to N) do
    Nb[i] ← i*i ;
  EndFor

  For (i ← 1 to N) do
    Write (Nb[i]) ;
  EndFor
End.
```

```
Algorithm Algo_2 ;
  Var i,N,S : integer ;
  Nb : array [50] of integer ;
Begin
  Repeat
    Read (N);
  Until ((N>0) and (N≤50)) ;

  S ← 0 ;
  For (i ← 1 to N) do
    Nb[i] ← i*i ;
    S ← S +Nb[i] ;
  EndFor

  Write (S) ;
End.
```

1- Translate the above algorithms into **C programs**.

2- Trace the execution of Program1 and Program2 step by step, and describe what does these programs.

Exercise 2

Let « A » an array of « N » real numbers (N does not exceed 120).

Write a **C program** to find and display the smallest value, largest value and their positions in the «A» array.

Exercise 3

Let « V » be an array of integers with size 60.

Write a **C program** that splits (divides) « V » into two arrays (V_POS and V_NEG) containing respectively the positive numbers and the strictly negative numbers of « V » array.

Exercise 4

Let « A » be an array of integers with size 80.

Write a **C program** that reverses the elements of « A » array.